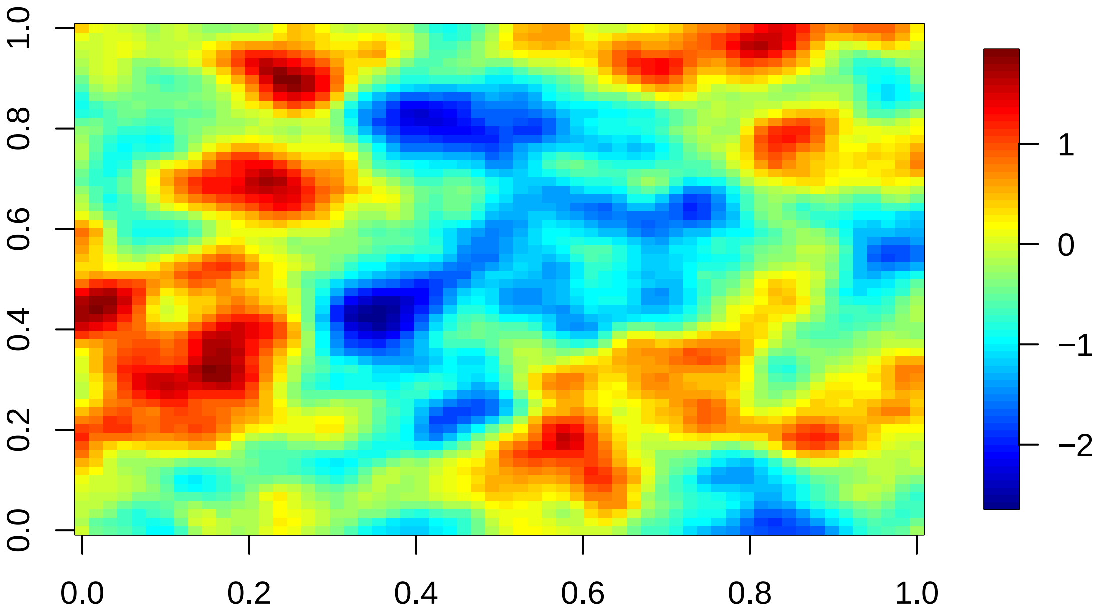
STAT 5845 Homework 7

November 10, 2025

Contents

1 Preliminaries

1.1 Simulate a Gaussian random field on a 60 × 60 grid from a Matern covariance



1.2 Randomly select 100 locations for training; use the remaining locations as testing.

```
# Select 100 Random Points as training points
n.train <- 100
train_idx <- sample.int(n.loc, n.train) # randomly select 100 integers from 1
                                        # to n = number of grid points

# Set the rest of the points in the grid to be used for kriging later
test_idx <- setdiff(seq_len(n.loc), train_idx)
train_locs <- locs[train_idx, , drop = FALSE]
train_Z <- Z[train_idx]
train_D <- as.matrix(dist(train_locs))
D_all_train <- fields::rdist(locs, train_locs)
```

2 Exponential

2.1 MLE Estimation Results & Check for Convergence

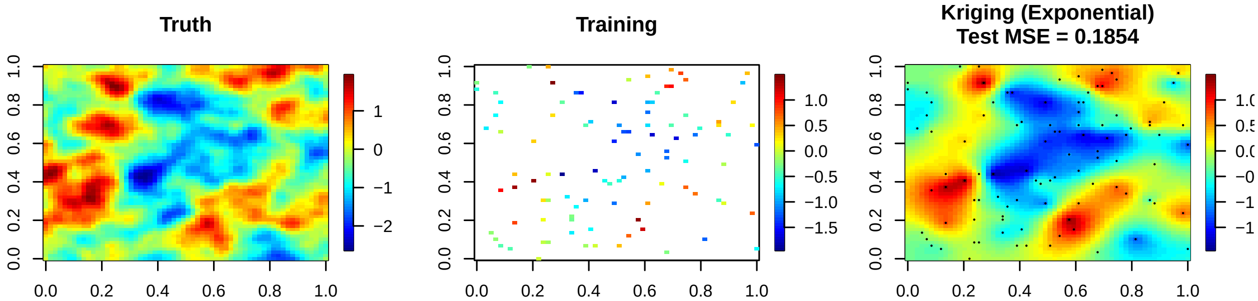
```
## List of 5
##  $ par         : num [1:4] -0.000028 0.790774 -1.877661 -0.198057
##  $ value       : num 3300
```

```
##  $ counts     : Named int [1:2] 185 NA
##   ..- attr(*, "names")= chr [1:2] "function" "gradient"
##  $ convergence: int 0
##  $ message    : NULL
```

```
## [1] "True vs. Estimated Parameters:"
```

```
##     Parameter True  Estimated
##         tau^2 0.00  7.854e-10
##       sigma^2 1.00  6.253e-01
##  beta (range) 0.05  1.529e-01
##     mu (mean) 0.00 -1.981e-01
```

2.2 Kriging & Corresponding Plot




```
## [1] "Exponential Covariance MSE: 0.1854"
```

---

3 Gaussian

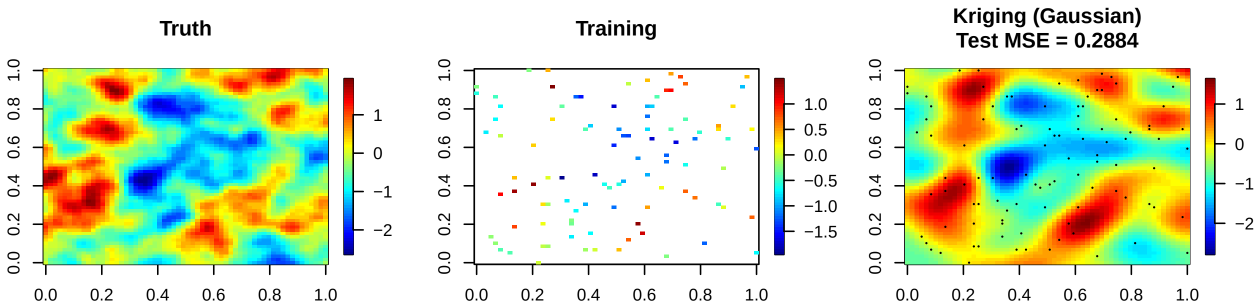
3.1 MLE Estimation Results & Check for Convergence

```
## List of 5
##  $ par        : num [1:4] 0.207 1.503 -1.866 1.838
##  $ value      : num 3319
##  $ counts     : Named int [1:2] 173 NA
##   ..- attr(*, "names")= chr [1:2] "function" "gradient"
##  $ convergence: int 0
##  $ message    : NULL
```

```
## [1] "True vs. Estimated Parameters:"
```

```
##     Parameter True Estimated
##         tau^2 0.00   0.04296
##       sigma^2 1.00   2.25942
##  beta (range) 0.05   0.15474
##     mu (mean) 0.00   1.83843
```

3.2 Kriging & Corresponding Plot




```
## [1] "Gaussian Covariance MSE: 0.2884"
```

---

4 Matern

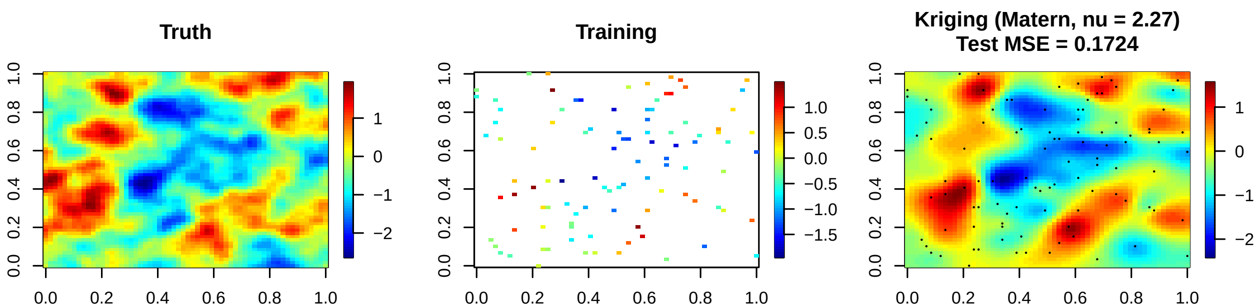
4.1 MLE Estimation Results & Check for Convergence

```
## List of 5
##  $ par        : num [1:5] -0.000121 0.807219 -3.203856 2.267827 -0.251631
##  $ value      : num 3291
##  $ counts     : Named int [1:2] 242 NA
##   ..- attr(*, "names")= chr [1:2] "function" "gradient"
##  $ convergence: int 0
##  $ message    : NULL
```

```
## [1] "True vs. Estimated Parameters:"
```

```
##         Parameter True  Estimated
##             tau^2 0.00  1.469e-08
##           sigma^2 1.00  6.516e-01
##      beta (range) 0.05  4.061e-02
##  nu (smoothness) 2.00  2.268e+00
##         mu (mean) 0.00 -2.516e-01
```

4.2 Kriging & Corresponding Plot




```
## [1] "Matern Covariance MSE: 0.1724"
```

5 Concluding Question: What do you observe about the kriging predictions under the Exponential, Gaussian, and Matern models? Which performs best based on the MSE, and why?

Based on the MLE estimates obtained here, the predictions made under the MLE-estimated Matern covariance function had the lowest MSE, followed closely by the MLE-approximated exponential covariance function. The Gaussian covariance MLE estimates did not prove to be very effective for this example, having a much higher MSE. Graphically, we observe an overly-smoothed prediction for the Gaussian MLE covariance function. While this covariance function had the worst predictions, we do see that the covariance function provided reasonable sensitivity for interpolation using kriging, though it missed some of the finer details of the trend.

The MLE-estimated exponential covariance function also had troubling predictions, missing finer details while capturing the general trends reasonably. Relatively, the MLE-esimated Matern covariance function did a bit of a better job at capturing some finer trends, but the fit had imperfections much like the other models.

Overall, we find that MLE estimation of covariance functions followed by kriging allows us to interpolate surfaces that make the "magic" of spatial prediction possible—a very valuable tool. However, we see that our predictions are limited by the effectiveness of maximum likelihood estimation for our covariance model parameters and the count of training data points given to interpolate with.

Special credit to Dr. Salvana for providing much of the relevant code for this assignment.